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Software Development Engineer II, AWS Redshift | Production AI Agents | Distributed Systems | Developer Infrastructure

SUMMARY

Software Development Engineer II at AWS Redshift building production AI-agent systems, large-scale distributed infrastructure, and developer automation used by hundreds of engineers. 5+ years across AWS, Zeta/Directi, and Ola/GeoSpoc, with ownership of Tier-1 cloud control-plane services, fleet-scale execution systems, GenAI approval workflows, region-build automation, observability, and secure operational tooling. Strong fit for applied AI engineering: agent orchestration, tool-using LLM systems, deterministic workflow state, human-in-the-loop gates, structured handoffs, eval/validation harnesses, production reliability, and long-horizon software engineering workflows.

CORE SKILLS

AI/Agents: multi-agent orchestration, production GenAI systems, LLM reasoning, tool-use workflows, LangGraph, agentic SDLC automation, context engineering, structured outputs, human-in-the-loop approval, workflow state machines, RAG/retrieval patterns, evals/validation, observability, safety guardrails

Backend/Cloud: Java, Python, Go, TypeScript, C/C++, distributed systems, microservices, orchestration, REST/GraphQL APIs, AWS CDK, ECS/Fargate, Lambda, Step Functions, SQS, DynamoDB, S3, CloudWatch, AppConfig, IAM, Docker, Kubernetes, CI/CD, security reviews

PROFESSIONAL EXPERIENCE

Software Development Engineer II | Amazon Web Services (Redshift) | Palo Alto, CA | Nov 2024 - Present

- Designed and shipped a production GenAI agent system for high-risk Redshift operational script review and approval, used by 200+ engineers to process 20+ risky scripts/day; automated 90% of approval decisions, saved 5-6 hours of manual review time per script, and achieved 92% reviewer agreement with AI-generated review outcomes.
- Combined large-scale language model reasoning, dynamic test execution, policy checks, context retrieval, and safe tool orchestration to solve manual approval gaps caused by missing real-time data, incomplete context, unsafe tool use, prompt injection risk, incomplete repository understanding, and long-running orchestration failures.
- Built a production-grade multi-agent SDLC orchestration platform used by 300+ engineers, with specialized agents coordinating requirements, design, implementation, review, validation, cleanup, and memory workflows; implemented deterministic workflow state, structured handoffs, context management, resume logic, operational recovery, human approval gates, and validation tooling.
- Architecting a ground-up redesign of a fleet-scale script execution platform from a partition-level monolithic service to a regional serverless architecture on ECS Fargate, reducing partition-wide blast-radius risk and replacing 300K+ manually managed scheduled jobs with a unified self-service execution model.
- Defined and drove cross-organization automation standards for Redshift region builds across 10+ engineering teams, delivering hundreds of automation tools, pipelines, testing frameworks, and operational workflows while leading architecture reviews, demos, coaching sessions, and adoption planning.
- Designed zero-touch region-build automation for multiple Redshift infrastructure components, eliminating 40+ manual steps and saving multiple weeks of engineering bandwidth per new region.
- Delivered reusable infrastructure to run Lambda-style logic on EC2 during early region build phases, bypassing Lambda availability dependencies and saving 2-3 weeks of build time per region.
- Built deployment status notifications across 6 core infrastructure pipelines, enabling automated progression through region-build stages and removing manual status checks and approval bottlenecks.
- Developed Unified Governance onboarding/offboarding APIs that enable customers to adopt Lakehouse architecture through a single API call instead of multiple manual steps across Redshift and Lake Formation.
- Improved operational safety by onboarding an internal proxy service to a self-service command framework, eliminating interactive access to customer accounts and reducing operator action time by 80%.
- Serve as a Security Guardian, conducting security reviews across and beyond Redshift to identify and remediate security gaps in projects and operational workflows.
- Led technical direction across 3 engineering teams of 4-5 engineers, authored 50+ design documents, mentored 20+ engineers directly, and onboarded newly transferred teams into Redshift engineering practices, architecture, and operational standards.

Software Development Engineer II | Amazon Web Services (Redshift) | Dublin, Ireland | Oct 2023 - Nov 2024

- Led development of a Tier-1 orchestration platform powering provisioning and lifecycle management for millions of Amazon Redshift customer instances each year, with ownership across design, implementation, testing, deployment, and operations.
- Architected and shipped Instance Configs, replacing a per-instance-type process that took up to 4 weeks with a centralized dynamic configuration system that deploys changes in under 1 hour and lets data-plane teams self-serve updates.
- Led cloud infrastructure setup for an internal operations console from the ground up using AWS CDK, provisioning networking, compute, and frontend hosting layers with a fully automated multi-region deployment pipeline.
- Built an end-to-end automated testing framework for the data collection platform, reducing manual deployment validation effort by 90% and supporting air-gapped government cloud regions.
- Built observability from scratch, including metrics pipelines, execution logs, and dashboards used in weekly reviews to detect ingestion anomalies and jobs negatively impacting customer clusters.
- Reduced infrastructure costs by 66% through modern instance migration and staging-environment restructuring; reduced memory consumption for a class of data collection jobs by 95%, projecting approximately $311K/year in commercial production savings.
- Onboarded the platform to three new ISO-compliant regions, cutting onboarding time from 2-3 weeks to 2-3 hours with infrastructure-as-code and automated workflows.

PROFESSIONAL EXPERIENCE CONTINUED

Software Development Engineer | Amazon Web Services (Redshift) | Dublin, Ireland | Sep 2022 - Sep 2023

- Designed and developed a mega-scale distributed script execution engine and backend services for Amazon Redshift, improving service reliability, operational efficiency, and customer experience across distributed cloud infrastructure.
- Built hundreds of automation frameworks to improve operational-standard compliance and minimize manual intervention for engineering and operations teams.
- Developed internal tooling used by engineers, managers, and on-call operators to monitor, diagnose, and manage Redshift infrastructure and service workflows, reducing operational friction and improving incident response.
- Implemented backend services, APIs, and workflow orchestration components for high-availability distributed systems running at AWS scale.

Software Development Engineer I | Zeta Suite (Directi) | Bengaluru, India | Jun 2021 - Sep 2022

- Built backend systems for enterprise core-banking functionality requiring strict availability and transactional consistency; developed APIs, enhanced platform functionality, debugged production issues, monitored system health, and participated in production deployments.
- Designed and developed an internal one-click deployment platform for provisioning a dynamic set of microservices on cloud infrastructure, compressing a 1-1.5 month manual deployment process to hours while reducing inter-team dependency bottlenecks, incident coordination, status tracking, and manual issue management.

Software Engineer - Full Stack | GeoSpoc, an Ola Company | Remote / Pune | Oct 2020 - Jun 2021

- Developed backend systems, database schemas, and robust APIs in Node.js/TypeScript; built reusable Angular UI components and SPAs; configured CI/CD pipelines; containerized applications with Docker for consistent deployment.

Software Developer - Full Stack | Exeliq Tech Solutions | Delhi NCR | Aug 2019 - Aug 2020

- Delivered MEAN-stack web applications with custom admin panels for content, image, SEO, and navigation management; implemented backend logic, APIs, MongoDB storage, and AWS EC2 production deployments.

SELECTED AI PLATFORM PROJECT

Production Multi-Agent SDLC Platform | Developer productivity and operational safety

- Created a multi-agent AI system used by 300+ engineers to coordinate specialized agents across software delivery workflows, including planning, design, implementation, review, infrastructure validation, cleanup, notifications, observability, and institutional memory.
- Implemented deterministic workflow routing, structured agent-to-agent handoffs, schema-validated outputs, context management, dependency handling, retry/circuit-breaker behavior, human-approval gates, and recovery logic to make long-running agent work auditable and resumable.
- Built asynchronous dispatch and monitoring infrastructure for long-running agent tasks, including run-state tracking, operational logging, failure recovery, prompt-injection-aware safeguards, and safe tool-use controls for production-style agent execution.
- Added validation and quality gates across agent behavior, handoff contracts, premise checks, review workflows, memory promotion, observability, and operational dispatch paths to improve reliability, reduce unsafe automation, and handle incomplete context or long-running orchestration failures.

EDUCATION & ACHIEVEMENTS

B.Tech., Electronics & Communication Engineering | Ajay Kumar Garg Engineering College, Ghaziabad | 80.3% | 2016 - 2020

National Gold Medal, World Robot Olympiad India 2018 | Rank 6, Robocon 2018 | Qualified: CodeVita 2019, HackWithInfy 2019